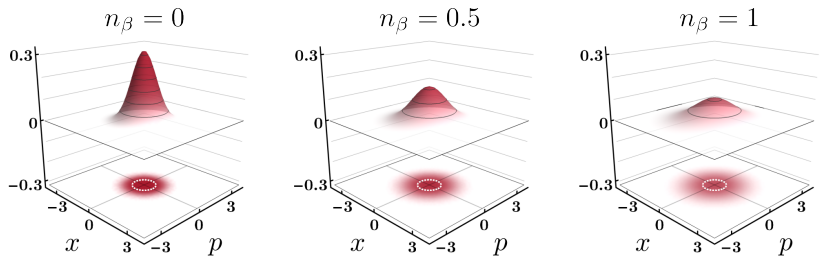

Figure 3.8: Thermal-state Wigner functions for $n_\beta = 0$ (the vacuum state), $n_\beta = 0, 5$, and $n_\beta = 1$. The dotted, white ring indicates one standard deviation of the vacuum distribution.

3.1.10 Thermal states

We conclude our general discussion of the quantum harmonic oscillator by introducing one more important class of states, which, unlike the states introduced so far, are not pure but mixed.

Imagine a general quantum system with arbitrary Hamiltonian $\hat{H}$ in contact with a large thermal bath with temperature T. If the system is in thermal equilibrium with the bath, then the energy is expected to follow a classical Boltzmann distribution (just like for any system in statistical mechanics). That is, the probability to find the system in an energy eigenstate with eigenvalue λ_n will be proportional to the so-called Boltzmann factor $e^{-\beta\lambda_n}$, where $\beta = 1/(k_B T)$ is the inverse temperature with k_B being the Boltzmann constant. There will be no coherences between eigenstates of different energy. The state of the system will therefore be diagonal in the energy eigenbasis and takes the form

$$\hat{\tau} = \frac{1}{Z}\sum_n e^{-\beta\lambda_n}|\lambda_n\rangle\langle\lambda_n| = \frac{1}{Z}e^{-\beta\hat{H}}, \tag{3.127}$$

where the normalisation constant $Z = \mathrm{Tr}[e^{-\beta\hat{H}}] = \sum_n e^{-\beta\lambda_n}$ is called the partition function. States of the form (3.127) are known as *Gibbs states*. They generally occur wherever quantum systems are in contact with thermal environments. Since in practice no system can be made perfectly closed, Gibbs states and more generally thermal noise are ubiquitous in implementations of quantum information technology across all the different physical platforms being used.

For harmonic oscillator systems (light in particular), the Hamiltonian is given by (3.22) and the eigenenergies (3.29) are equally spaced. The Gibbs state (3.127) becomes

$$\hat{\tau}_\beta = \frac{e^{-\beta\hbar\omega(\hat{n}+\frac{1}{2})}}{\mathrm{Tr}[e^{-\beta\hbar\omega(\hat{n}+\frac{1}{2})}]} = (1 - e^{-\beta\hbar\omega})\sum_{n=0}^{\infty} e^{-\beta\hbar\omega n}|n\rangle\langle n|, \tag{3.128}$$

and is then known as a *thermal state*. The mean excitation number in the thermal state is

$$n_\beta = \mathrm{Tr}[\hat{\tau}_\beta\hat{n}] = \frac{1}{e^{\beta\hbar\omega} - 1}. \tag{3.129}$$

The thermal state is often parametrised in terms of this mean excitation number rather than the (inverse) temperature as

$$\hat{\tau}_\beta = \frac{1}{n_\beta + 1} \sum_{n=0}^{\infty} \left(\frac{n_\beta}{n_\beta + 1} \right)^n |n\rangle\langle n|. \tag{3.130}$$

What does the thermal state look like in phase space? Since $\hat{\tau}_\beta$ is diagonal in the Fock basis and all the Fock states are rotationally symmetric in phase space, it is clear that $\hat{\tau}_\beta$ will also be symmetric. The Wigner function is

$$W_{\tau_\beta}(x,p) = \frac{1}{\pi(1+2n_\beta)} e^{-\frac{1}{1+2n_\beta}(x^2+p^2)}. \tag{3.131}$$

Like the coherent and squeezed states, the thermal states are Gaussian states. In fact, they are similar to vacuum states with the quadrature variances increased from $\frac{1}{2}$ to $\langle \Delta^2 \hat{x} \rangle = \langle \Delta^2 \hat{p} \rangle = \frac{1}{2} + n_\beta$. Fig. 3.8 shows plots of thermal-state Wigner functions. Note how the width of the distribution grows with n_β.